

Notes: La Porta & López-de-Silanes (QJE, 1999)

The Benefits of Privatization: Evidence from Mexico

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	2
4	Models and empirical specifications	2
4.1	Before-after and industry-adjusted comparisons	2
4.2	Decomposition logic	2
4.3	Cross-sectional regressions	2
5	Identification and empirical design	3
6	Tables and figures	3
6.1	Table I: Privatization Contracts According to Industry Classification; Table II: Observations in the Sample	3
6.2	Table III: Changes in Performance; Table IV: Industry-Adjusted Changes in Performance . .	4
6.3	Figures I–III: Employment, Operating Income/Sales, and Cost per Unit	4
6.4	Table V: What Explains Increased Profitability?; Table VI: Competitive versus Noncompetitive Industries	5
6.5	Table VII: Transfers from Workers; Table VIII: Compensation of Workers	6
6.6	Table IX: Regression Results	7
7	Short summary	7
8	Conclusion	8
8.1	Core conclusion	8
8.2	Incremental lesson	8
8.3	Policy implication	8
8.4	Takeaway	8

1 Big picture

- **Question.** Did Mexican privatization improve firm performance, and if profitability rose, did gains come from productivity, prices, or worker transfers?
- **Setting.** Mexico privatized firms across many industries between 1983 and 1991.
- **Main result.** Privatized firms become much more profitable and close the gap with industry-matched private firms.
- **Channels.** Higher prices explain little; worker transfers explain a meaningful share; productivity and cost reductions explain most of the gain.
- **Takeaway.** Privatization generated efficiency gains, but employment losses created important distributional costs.

2 Incremental contribution of the paper

- Near-census evidence for Mexican nonfinancial privatizations.
- Direct decomposition of profit gains into prices, worker transfers, and productivity.
- Product-level price and quantity data plus detailed wage and employment data.
- Industry-matched private control groups to separate privatization from industry trends.
- Explicit comparison of social, managerial-agency, and political-agency views of SOEs.

3 Data and measurement

The sample includes 218 nonfinancial privatization contracts, with usable data for 170 firms accounting for more than 97 percent of sales and assets of privatized nonfinancial firms. Outcomes include profitability, efficiency, labor, investment, output, prices, and taxes. Main comparisons use before/after and industry-adjusted changes.

$$AdjOutcome_{i,t} = Outcome_{i,t} - BenchmarkIndustryOutcome_{j(i),t}.$$

Interpretation: Industry adjustment compares each SOE to private firms in its three-digit industry, reducing confounding from macro and industry trends.

4 Models and empirical specifications

4.1 Before-after and industry-adjusted comparisons

$$Change_i = Outcome_{i,post} - Outcome_{i,pre}, \quad AdjChange_i = AdjOutcome_{i,post} - AdjOutcome_{i,pre}.$$

4.2 Decomposition logic

$$Profitability\ gain = price\ effect + worker\ transfer + productivity/efficiency\ effect + residual.$$

4.3 Cross-sectional regressions

$$AdjChange_i = \alpha + \beta_1 Monopoly_i + \beta_2 PriceControlled_i + \beta_3 FDIRestricted_i + \beta_4 RevenuePrivatization_i + \beta_5 OutsideBuyer_i + \beta_6$$

5 Identification and empirical design

The paper is not a randomized experiment. It combines comprehensive sampling, before-after comparisons, industry-matched controls, product-price evidence, labor/wage evidence, survey evidence, and cross-sectional regressions. The strongest identification claim is about the *composition* of performance changes: price increases, worker transfers, and productivity gains can be measured directly. The main threat is that privatization timing and selection may be correlated with unobserved reform potential.

6 Tables and figures

6.1 Table I: Privatization Contracts According to Industry Classification; Table II: Observations in the Sample

Read: The first table shows the industry spread of contracts; the second shows that the usable sample covers over 97 percent of sales and assets.

Economic message: The sample is close to the universe of Mexican nonfinancial privatizations, increasing external validity.

TABLE I
PRIVATIZATION CONTRACTS ACCORDING TO INDUSTRY CLASSIFICATION

Industry	Number of contracts	Percentage of assets of the sample	Percentage of labor of the sample
<i>I. Mining</i>			
Mining of metallic minerals	6	0.0853	0.0389
Mining of nonmetallic minerals	6	0.0083	0.0105
<i>II. Manufacturing</i>			
Milk products	4	0.0015	0.0145
Fruits and vegetables	4	0.0003	0.0025
Canned fish and seafood	12	0.0060	0.0191
Grains and oils	9	0.0105	0.0209
Sugar mills	44	0.0510	0.1699
Animal foods	8	0.0016	0.0034
Beverages	2	0.0005	0.0033
Tobacco products	3	0.0020	0.0110
Textiles, clothing, and leather	7	0.0061	0.0145
Wood and wood products	2	0.0005	0.0020
Paper and printing	5	0.0104	0.0241
Chemicals, oil products, and plastic	23	0.0659	0.0376
Nonmetallic mineral products	9	0.0084	0.0145
Basic metals and derivative products	22	0.2224	0.1672
Heavy machinery and equipment	7	0.0040	0.0086
Electronics, machinery, and equipment	6	0.0130	0.0292
Automotive industry	15	0.0284	0.0469
Transportation equipment	2	0.0028	0.0109
<i>III. Services</i>			
Hotels and restaurants	8	0.0054	0.0231
Land and sea transportation	4	0.0233	0.0021
Air transportation	3	0.0819	0.1053
Communications	1	0.3590	0.2178
Real estate and professional services	4	0.0017	0.0018
Recreation and entertainment services	2	0.0000	0.0004
Total	218	1.00	1.00

This table classifies privatization contracts across industries. The table indicates for each industry (1) the number of contracts signed, (2) the ratio of the preprivatization value of the industry's total assets to the total preprivatization value of all the assets in the sample, and (3) the ratio of the industry's preprivatization total employment to the total preprivatization employment of all the firms in the sample.

(a) Table I: industry distribution of privatization contracts.

TABLE II
OBSERVATIONS IN THE SAMPLE

	Number	Percentage of sales	Percentage of assets	Percentage of employment
Firms in our sample	170	97.31	97.17	92.10
Merged and no longer keep independent accounting records	9	0.17	0.35	0.48
Firms not found due to incomplete census records, name changes, etc.	26	1.69	1.56	5.22
Liquidated firms	13	0.83	0.92	2.20
All privatized firms (1983–1991)	218	100.00	100.00	100.00

This table breaks all nonfinancial firms privatized in Mexico between 1983 and 1991 into four groups. For each group we provide (1) the number of firms, (2) the percentage of preprivatization sales in the total, (3) the percentage of preprivatization assets in the total, and (4) the percentage of preprivatization employment in the total.

(b) Table II: observations in the sample.

6.2 Table III: Changes in Performance; Table IV: Industry-Adjusted Changes in Performance

Read: Profitability and efficiency rise, employment falls, and tax payments rise. Industry adjustment preserves the main efficiency gains.

Economic message: The raw and adjusted tables show real restructuring, not merely economy-wide trends.

TABLE III
CHANGES IN PERFORMANCE FOR THE SAMPLE OF PRIVATIZED FIRMS

Variable	N	Mean before Median before	Mean after Median after	T-statistic for change in mean	Z-statistic for change in median
<i>I. Profitability</i>					
Operating income/sales	170	-0.1546 -0.0273	0.0865 0.0935	5.43 ^a	5.62 ^a
Operating income/PPE	170	-0.1053 -0.0142	0.2397 0.1305	4.38 ^a	5.18 ^a
Net income/sales	170	-0.3632 -0.0142	0.0364 0.1305	4.84 ^a	6.48 ^a
Net income/PPE	170	-0.2176 -0.0728	0.0537 0.0839	2.59 ^a	5.82 ^a
<i>II. Operating efficiency</i>					
Cost per unit	170	0.9166 0.8805	0.7017 0.7129	-6.68 ^a	-6.95 ^a
Log (sales/PPE)	170	0.0570 0.1680	0.7034 0.4065	4.25 ^a	3.64 ^a
Log (sales/employees)	169	4.3674 4.2054	5.4204 5.1963	8.33 ^a	8.35 ^a
Operating income/employees	169	1.6709 -0.6826	54.1750 14.4131	6.36 ^a	5.36 ^a
<i>III. Labor</i>					
Log (employees)	169	6.1655 6.2976	5.5166 5.7301	-4.37 ^a	-4.35 ^a
Log (blue-collar workers)	168	5.8222 5.9558	5.2878 5.3471	-3.62 ^a	-3.62 ^a
Log (white-collar workers)	169	4.8367 4.7875	4.3015 4.3241	-4.17 ^a	-4.36 ^a
<i>IV. Assets and investment</i>					
Log (PPE)	170	10.5194 10.4582	10.2395 10.5497	-1.17	0.34
Investment/sales	170	0.0299 0.0022	0.0449 0.0125	1.77 ^c	4.13 ^a
Investment/employees	169	3.1655 0.1335	10.3074 2.7184	4.69 ^a	5.96 ^a
Investment/PPE	170	0.0244 0.0040	0.0718 0.0256	4.50 ^a	5.36 ^a
Log (PPE/employees)	169	4.4645 4.2978	4.6624 4.8014	1.15	1.86 ^c
<i>V. Output</i>					
Log (sales)	170	10.4746 10.4168	11.0174 11.0984	3.25 ^a	3.74 ^a
<i>VI. Prices</i>					
Index of real prices (Paasche)	83	100.00 100.00	101.31 101.27	0.31	0.72
<i>VII. Net taxes</i>					
Net taxes/sales	170	-0.0462 0.0000	0.0839 0.0763	6.95 ^a	10.80 ^a
Net taxes	170	-760.24 0.0000	25,684.12 2,162.80	2.21 ^b	10.94 ^a

This table presents raw results for the sample of 170 privatized firms. The table presents, for each empirical proxy, the number of usable observations, the mean, and the median values before privatization and after privatization (in 1993). We report *t*-statistics and *z*-statistics (Wilcoxon rank sum) as our test for significance for the change in mean and median values, respectively. Definitions for each variable can be found in the Appendix.

^a Significant at 1 percent

(a) Table III: raw before/after changes.

TABLE IV
INDUSTRY-ADJUSTED CHANGES IN PERFORMANCE FOR THE SAMPLE OF PRIVATIZED FIRMS

Variable	N	Mean before Median before	Mean after Median after	T-statistic for change in mean	Z-statistic for change in median
<i>I. Profitability</i>					
Operating income/sales	168	-0.2975 -0.1353	0.0289 0.0178	4.35 ^a	7.15 ^a
Operating income/PPE	168	-0.2336 -0.1471	0.0914 0.0021	4.12 ^a	4.87 ^a
Net income/sales	168	-0.4036 -0.1720	0.0108 0.0401	4.69 ^a	6.99 ^a
Net income/PPE	168	-0.2686 -0.1494	-0.0162 0.0242	2.41 ^a	5.76 ^a
<i>II. Operating efficiency</i>					
Cost per unit	168	0.1827 0.1423	-0.0021 -0.0105	-5.68 ^a	-5.63 ^a
Log (sales/employees)	166	-1.1836 -1.3133	-0.2679 -0.4299	6.47 ^a	6.85 ^a
Log (sales/PPE)	168	-0.2850 -0.1640	0.1834 0.0374	3.11 ^a	2.35 ^b
<i>III. Labor</i>					
Index of total employees	169	100.00 100.00	64.65 65.63	-12.48 ^a	-9.29 ^a
Index of blue-collar workers	168	100.00 100.00	63.22 67.39	-11.73 ^a	-9.08 ^a
Index of white-collar workers	169	100.00 100.00	67.79 68.41	-11.12 ^a	-7.05 ^a
<i>IV. Assets and investment</i>					
Investment/sales	168	0.0116 -0.0085	-0.0358 -0.0580	-5.41 ^a	-10.26 ^a
Investment/PPE	168	-0.0037 -0.0127	-0.0365 -0.0630	-2.89 ^a	-7.76 ^a
<i>V. Output</i>					
Log (sales)	170	-2.1189 -2.2166	-1.6865 -1.7275	1.91 ^a	2.70 ^a
<i>VI. Net taxes</i>					
Net taxes/sales	168	-0.0944 -0.0168	0.0665 0.0589	4.08 ^a	10.18 ^a
Net taxes	168	-766.70 -8.5112	25,522.28 2,079.71	2.19 ^b	10.53 ^a

This table presents industry-adjusted results for the sample of 170 privatized firms. We form (three-digit S.I.C. code level) industry control groups using all private firms trading in the Mexican Stock Market. For each privatized SOE and for each year, we compute industry-adjusted indicators by taking the difference between the value of the indicator for the SOE and its industry control group. We use economywide aggregates, if available, for those firms for which we cannot find a matched industry sample. The Appendix provides more details for each individual measure. The table presents, for each empirical proxy, the number of usable observations, the mean, and the median values before privatization and after privatization (in 1993). We report *t*-statistics and *z*-statistics (Wilcoxon rank sum) as our test for significance for the change in mean and median values, respectively. Definitions for each variable can be found in the Appendix.

(b) Table IV: industry-adjusted changes.

6.3 Figures I–III: Employment, Operating Income/Sales, and Cost per Unit

Read: Employment falls after privatization, while operating income/sales rises and cost per unit falls.

Economic message: The figures summarize the restructuring pattern: fewer workers, higher profitability, and lower costs.

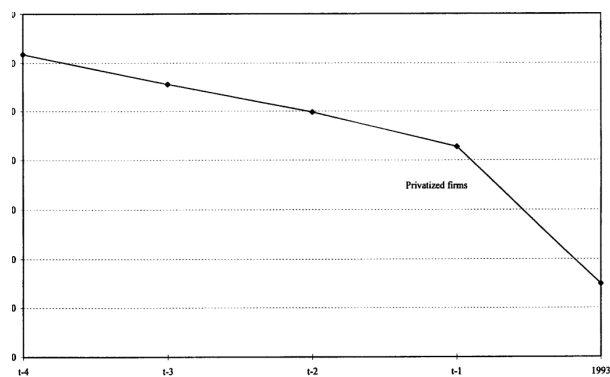


FIGURE I
Mean Total Employment of Privatized Firms

(a) Figure I: employment path.

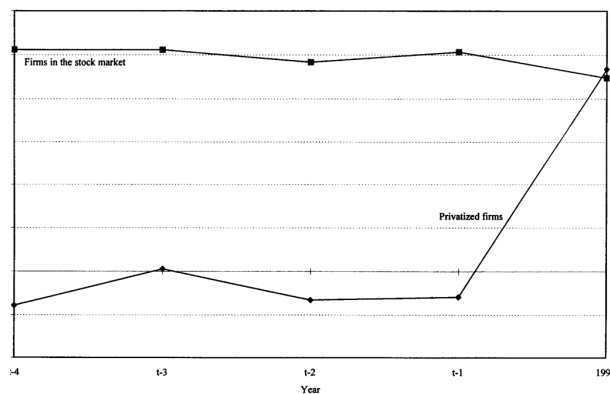


FIGURE II
Firms in the Stock Market versus Privatized Firms

(b) Figure II: operating income/sales.

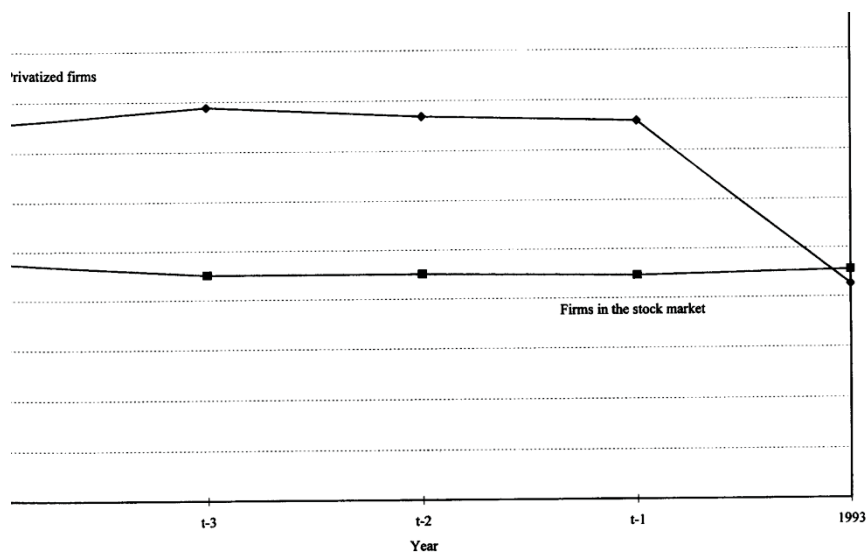


FIGURE III
Firms in the Stock Market versus Privatized Firms
Median Cost per Unit

Figure III: cost per unit.

6.4 Table V: What Explains Increased Profitability?; Table VI: Competitive versus Noncompetitive Industries

Read: Survey responses point to new processes, labor changes, product changes, and access to capital. Competitive and noncompetitive industries both improve.

Economic message: Productivity and restructuring, not pure monopoly pricing, dominate the explanation.

Reasons explaining increased performance:	N	Percent of respondents assigning:				T-statistics of the test of difference in means between:		
		Maximum score	Minimum score	Median	Mean	Standard error	The next reason	Two reasons below
tion of new production processes	74	50.00	4.05	4.5	4.2162	0.1206	0.67 (1 vs. 2)	1.96 ^a (1 vs. 3)
d managers and directors	74	43.24	1.35	4	4.0946	0.1204	1.12 (2 vs. 3)	0.88 (2 vs. 4)
in the organization of labor	74	35.14	2.70	4	3.8919	0.1241	-0.49 (3 vs. 4)	1.37 (3 vs. 5)
the labor force	74	31.08	1.35	4	3.9595	0.1026	1.96 ^a (4 vs. 5)	4.48 ^a (4 vs. 6)
ments in the quality of existing	74	25.68	6.76	4	3.6487	0.1339	3.03 ^a (5 vs. 6)	3.69 ^a (5 vs. 7)
ts	74	22.97	21.62	3	3.0405	0.1715	0.50 (6 vs. 7)	1.24 (6 vs. 8)
capital markets	74	22.97	21.62	3	2.9460	0.1720	0.94 (7 vs. 8)	-0.39 (7 vs. 9)
st of inputs	74	17.57	22.97	3	2.7703	0.1638	-1.16 (8 vs. 9)	1.90 ^a (8 vs. 10)
tion of new products	74	12.16	20.27	3	3.0270	0.1551	2.96 ^a (9 vs. 10)	5.45 ^a (9 vs. 11)
in the distribution network	74	9.46	39.19	2	2.4189	0.1598	2.19 ^a (10 vs. 11)	-1.01 (10 vs. 12)
uring of suppliers	74	4.05	37.84	2	2.0270	0.1201	-3.77 ^a (11 vs. 12)	—
in the collective labor contract	74	2.70	20.27	3	2.5946	0.1318	—	—
d incentives for managers and	74							
rs	74							

a faced questionnaire sent to a random sample of over half (60) of the 170 privatized firms. The questionnaire was answered by either the CEO or the CFO of the firm. This table tabulates the responses to the question in the questionnaire which read: "Please indicate the relative importance of the following concepts in explaining the ability and performance of your firm after privatization." The respondents were asked to rate on a scale from 1 (not important) to 5 (very important) the relative importance of each concept. The different reasons that explain profitability are shown ranked according to the percentage of respondents assigning each category the most. Respondents could also answer "do not know" if they did not have enough information to provide an answer or provide alternative explanations. Only four of reasons were given, each by a different firm: (1) continuity in strategy, (2) increased productivity, (3) employees' commitment to the firm, and (4) independent technology.

at 1 percent.

at 5 percent.

at 10 percent.

(a) Table V: survey explanations.

Variable	Competitive according to prospectus				Noncompetitive according to prospectus				Competitive vs. noncompetitive			
	N	Median before	Median after	Median change	N	Median before	Median after	Median change	Difference in median change	Z-stat of difference in median change		
ability	134	-0.0537	0.0884	0.1457 ^a	32	0.0397	0.1306	0.0845 ^a	0.0612	0.30		
rating income/sales	133	-0.0396	0.1265	0.1745 ^a	32	0.0509	0.1460	0.1382 ^a	0.0363	0.75		
rating income/PPE	131	-0.1449	0.0635	0.2078 ^a	32	-0.0518	0.1257	0.3540 ^a	-0.1462	-1.37		
income/PPE	131	-0.1036	0.0809	0.2155 ^a	32	-0.0198	0.1127	0.2482 ^a	-0.0327	-0.86		
ating efficiency	134	0.8878	0.7244	-0.1296 ^a	32	0.8291	0.6130	-0.2362 ^a	0.1066	2.52 ^a		
t per unit	136	4.1798	5.1461	0.9858 ^a	32	4.5099	5.9394	0.8353 ^a	0.1505	0.75		
(sales/employees)	134	0.2186	0.6075	0.5132 ^a	32	-0.3804	-0.0382	0.3810	0.1322	0.22		
(sales/PPE)	135	-1.5385	14.398	19.161 ^a	33	6.8468	24.812	30.989 ^a	-11.828	-0.93		
rating income/employees	136	6.3122	5.6699	-0.4592 ^a	33	6.2260	5.9349	-0.1864	-0.2728	-2.09 ^a		
or	136	6.0235	5.3706	-0.4365 ^a	33	5.6348	5.3471	-0.2339	-0.2026	-1.50		
(employees)	136	4.7095	4.2905	-0.4443 ^a	33	5.1998	4.9715	-0.1540	-0.2903	-2.77 ^a		
(blue-collar workers)	136	6.0235	5.3706	-0.4365 ^a	33	5.6348	5.3471	-0.2339	-0.2026	-1.50		
(white-collar workers)	136	4.7095	4.2905	-0.4443 ^a	33	5.1998	4.9715	-0.1540	-0.2903	-2.77 ^a		
ts and investment	134	10.365	10.214	-0.0755	32	10.812	11.486	0.2342	-0.3097	-1.65 ^a		
(PPE)	133	0.0013	0.0094	0.0055 ^a	32	0.0099	0.0247	0.0109	-0.0054	-0.48		
stment/sales	133	0.0017	0.0217	0.0144 ^a	32	0.0107	0.0374	0.0180 ^a	-0.0036	-0.57		
stment/PPE	135	4.2471	4.7520	0.2874	33	4.4545	5.4488	0.4939	-0.2065	-1.08		
(PPE/employees)	135	10.392	10.968	0.5622 ^a	33	10.858	11.751	0.7776 ^a	-0.2154	-1.49		
ut	135	10.392	10.968	0.5622 ^a	33	10.858	11.751	0.7776 ^a	-0.2154	-1.49		
(sales)	135	10.392	10.968	0.5622 ^a	33	10.858	11.751	0.7776 ^a	-0.2154	-1.49		
rs	135	10.392	10.968	0.5622 ^a	33	10.858	11.751	0.7776 ^a	-0.2154	-1.49		
Price index (Paische)	67	100.00	99.72	-0.2768	16	100.00	105.94	5.9355 ^a	-6.2123	-1.74 ^a		
taxes/sales	133	0.0000	0.0586	0.0668 ^a	33	0.0000	0.0992	0.0940 ^a	-0.0272	-1.46		
taxes	135	0.0000	1.603.3	1.922.4 ^a	33	0.0000	6.626.3	8.947.8 ^a	-7.025.4	-3.05 ^a		

able presents raw performance results for the sample of 170 privatized firms. The table breaks down performance results into competitive and noncompetitive sectors based on the firm's industry provided by the financial consultants in the privatization prospectus of the firm. For each group of firms, the table presents median values before and after on and the z-statistics for the test of change in median. The last two columns of the table show the difference in median change across groups and the corresponding z-statistic.

(b) Table VI: competitive vs. noncompetitive industries.

6.5 Table VII: Transfers from Workers; Table VIII: Compensation of Workers

Read: Employment falls, real wages for remaining workers rise, and total wages/sales falls.

Economic message: Workers as a group bear part of the cost through layoffs, while remaining workers often receive higher wages.

Variable	N	Mean before (median)	Mean in 1993 (median)	t-stat for change (Z-test)
<i>Panel A: All firms with available wage data</i>				
Operating income/sales	101	-0.1530	0.0682	4.23 ^a
		-0.0251	0.0708	4.16 ^a
Real wages per worker	101	NS\$15,074	NS\$27,079	5.42 ^a
		NS\$9,995	NS\$22,481	6.05 ^a
Index of industry-adjusted real wages per worker	101	100.00	209.30	8.32 ^a
		100.00	198.96	6.47 ^a
Real wages per blue-collar worker	101	NS\$9,525	NS\$22,425	5.72 ^a
		NS\$6,791	NS\$16,851	7.45 ^a
Index of industry-adjusted real blue-collar wages	101	100.00	265.61	8.76 ^a
		100.00	222.43	7.04 ^a
Real wages per white-collar worker	101	NS\$27,992	NS\$44,301	4.13 ^a
		NS\$18,892	NS\$37,964	5.17 ^a
Index of industry-adjusted white-collar wages	101	100.00	177.97	6.08 ^a
		100.00	147.92	4.84 ^a
Total wages/sales	101	0.2338	0.1441	-3.14 ^a
		0.1506	0.1143	-2.93 ^a
Log (employees)	101	6.4405	5.8944	-2.95 ^a
		6.4708	5.9269	-2.99 ^a
<i>Panel B: Firms with available wage data that increased the number of workers after privatization</i>				
Operating income/sales	20	-0.2099	0.0342	2.17 ^b
		-0.0468	0.0952	2.08 ^b
Real wages per worker	20	NS\$12,727	NS\$26,203	2.32 ^b
		NS\$9,473	NS\$19,936	2.62 ^a
Real wages per blue-collar worker	20	NS\$8,679	NS\$19,608	2.65 ^a
		NS\$6,834	NS\$14,361	3.28 ^a
Real wages per white-collar worker	20	NS\$22,003	NS\$52,038	2.06 ^b
		NS\$15,010	NS\$37,602	3.25 ^a
Total wages/sales	20	0.2892	0.2121	-0.82 ^a
		0.1438	0.1702	0.03
Log (employees)	20	6.2785	6.6788	0.98
		6.3966	6.7440	1.76 ^c

The table presents data for all workers, blue-collar, and white-collar workers and shows real wages per worker and the index of industry-adjusted real wages per worker for all three groups. The table is divided into two panels. Panel A contains statistics for all 101 firms with available wage data. Panel B describes the wage behavior for the subsample of firms that increased the number of employees after privatization. We report t-statistics and z-statistics (Wilcoxon rank sum) as our test for significance for the change in mean and median values, respectively. Definitions for each variable can be found in the Appendix.

(a) Table VII: worker transfers.

Reasons for increased total compensation of workers:	N	Percent of respondents assigning:				t-statistics of the test of difference in means between:		
		Maximum score	Minimum score	Median	Mean	Standard error	The next reason	Two reasons below
ased productivity	59	74.58	0.00	5	4.6949	0.0686	4.91 ^a (1 vs. 2)	5.06 ^a (1 vs. 3)
ved supervision of workers	59	47.46	0.00	4	4.0000	0.1276	-0.27 (2 vs. 3)	0.41 (2 vs. 4)
mployees	59	39.90	5.08	4	4.0339	0.1094	0.64 (3 vs. 4)	9.27 ^a (3 vs. 5)
g more qualified workers and	59	33.90	0.00	4	3.9153	0.1261	6.02 ^a (4 vs. 5)	—
vees	59	33.90	0.00	4	3.9153	0.1261	6.02 ^a (4 vs. 5)	—
the conditions offered by	59	33.90	0.00	4	3.9153	0.1261	6.02 ^a (4 vs. 5)	—
at firms	59	8.47	42.37	2	2.3729	0.1541	—	—

i from a faced questionnaire sent to a random sample of over half (60) of the 170 privatized firms. The questionnaire was answered by either the CEO or the CFO of the firm. This table tabulates the responses to the question in the questionnaire which read: "Please indicate the relative importance of the following concepts in explaining the total compensation of workers in your firm after privatization." The respondents were asked to rate on a scale from 1 (not important) to 5 (very important) the relative importance of each concept. Only two of such alternative reasons were given, each by a different firm: (1) the existence of a "productivity contract" with the workers, and (2) flexible wages. The table is the answers of those firms that had increased both total compensation of blue-collar and of white-collar employees after privatization. The different reasons that explain total compensation of workers after privatization are shown ranked according to the percentage of respondents assigning each category the maximum score possible.

least at 1 percent.

least at 5 percent.

least at 10 percent.

(b) Table VIII: compensation explanations.

6.6 Table IX: Regression Results

Read: Cross-sectional regressions show that market-structure and deregulation variables have mixed effects, while preprivatization performance strongly predicts catch-up.

Economic message: The regressions reinforce the view that restructuring and catch-up dominate pure market-power explanations.

TABLE IX
REGRESSION RESULTS

<i>Independent variables:</i>	<i>Dependent variables:</i>						
	Adj Δ IO/S	Adj Δ NI/S	Adj Δ C/S	Adj Δ S/L	Adj Δ L	Adj I/PPE	Adj Δ S
Monopoly or oligopoly	0.0240 (0.0623)	0.0377 (0.0912)	0.0321 (0.0476)	-0.2023 (0.2474)	0.0987 ^b (0.0491)	-0.0184 (0.0295)	0.7583 ^a (0.2455)
Price/quantity controlled	-0.1154 ^b (0.0499)	-0.2126 ^a (0.0673)	0.1679 ^a (0.0419)	-0.3347 ^b (0.1707)	0.0661 ^c (0.0395)	-0.0385 ^b (0.0186)	0.4368 ^a (0.1665)
FDI restricted	0.0611 (0.0534)	0.0178 (0.0750)	-0.0591 (0.0399)	0.1799 (0.2056)	-0.0364 (0.0378)	0.0818 ^b (0.0380)	-0.1409 (0.2020)
Revenue privatization	0.0415 (0.0690)	0.0381 (0.1036)	0.0149 (0.0589)	-0.1038 (0.3481)	-0.0643 (0.0766)	0.0348 (0.0500)	-0.0554 (0.3288)
Outside buyer	0.0085 (0.1773)	0.0961 (0.2622)	0.0089 (0.1281)	-0.6394 ^c (0.3710)	-0.0333 (0.0708)	-0.0345 (0.0436)	-0.7967 ^b (0.3980)
Nontraded good	0.0750 (0.0929)	0.0436 (0.1180)	-0.1144 ^c (0.0616)	-0.1071 (0.2804)	0.0549 (0.0419)	0.0354 (0.0386)	0.1997 (0.2543)
Number of strikes	-0.1961 (0.0149)	-0.0378 ^b (0.0188)	0.0106 (0.0096)	-0.0405 (0.0417)	0.0021 (0.0114)	-0.0030 (0.0068)	-0.0215 (0.0526)
Adj OI/S _{pre}	-1.0658 ^a (0.0613)	—	—	—	—	—	—
Adj NI/S _{pre}	—	-0.9978 ^a (0.0620)	—	—	—	—	—
Adj RC/S _{pre}	—	—	-0.9607 ^a (0.0764)	—	—	—	—
Adj S/L _{pre}	—	—	—	-0.3920 ^a (0.0806)	—	—	—
Adj Δ L _{pre}	—	—	—	—	-1.0872 ^b (0.4385)	—	—
I/PPE _{pre}	—	—	—	—	—	-0.3652 (0.3244)	—
Adj sales _{pre}	—	—	—	—	—	—	-0.1702 ^a (0.0547)
Constant	0.0595 ^a (0.0208)	0.0665 ^a (0.0206)	0.0055 (0.0307)	1.7247 ^a (0.0928)	-0.4564 ^a (0.0313)	-0.0269 ^a (0.0082)	-0.0835 (0.0541)
Number of observations	163	160	163	138	164	162	165
Adj. R ²	0.5926	0.6179	0.4673	0.1700	0.2039	0.1143	0.1665

The table reports results from OLS regressions for the cross section of 170 privatized firms. All regressions include—but we do not report—eight dummies for privatization cohorts. White [1980] error-corrected standard errors are reported in parentheses. The dependent variables are (1) Adj Δ IO/S—the change in the industry-adjusted operating-income-to-sales ratio, (2) Adj Δ NI/S—the change in the industry-adjusted net-income-to-sales ratio, (3) Adj Δ C/S—the change in the industry-adjusted costs-per-unit, (4) Adj Δ S/L—the change in the industry-adjusted sales-to-labor ratio, (5) Adj Δ L—the growth in employment relative to the industry, (6) Adj I/PPE—the industry-adjusted investment-to-fixed-assets ratio, and (7) Adj Δ S—the industry-adjusted growth in sales. The independent variables are (1) Monopoly or oligopoly—a dummy variable that takes the value of one if the prospectus describes the industry in which the firm operates as monopolistic or oligopolistic and zero otherwise, (2) Price/quantity controlled—a dummy that takes the value of one if price of the firm's output or its quantity is controlled by the government and zero otherwise, (3) FDI restricted—a dummy variable that takes the value of one if foreigners are not allowed to own majority stakes in the sector in which the firm operates and zero otherwise, (4) Revenue privatization—a dummy that takes the value of one if the state owned a noncontrol package before the transaction took place and zero otherwise; (5) Outside buyer—a dummy variable that takes the value of one if the SOE is sold to a buyer with no experience in the same three-digit level S.I.C. classification and zero otherwise, (6) Nontraded good—a dummy variable that takes the value of one if it is not possible to trade the firm's output internally and zero otherwise, (7) Number of

Table IX: regression results.

7 Short summary

- Profitability and efficiency rise sharply after privatization.
- Higher prices explain only a small share of the gain.
- Employment falls, and worker transfers explain a meaningful share of the gain.
- Productivity gains account for most of the remaining improvement.

- Improvements occur in competitive and noncompetitive industries.

8 Conclusion

8.1 Core conclusion

Privatization in Mexico generated large efficiency gains, but also large labor-market distributional effects.

8.2 Incremental lesson

The paper's key contribution is channel decomposition: it distinguishes price extraction, labor transfers, and productivity gains.

8.3 Policy implication

Privatization can improve efficiency, but transition costs for displaced workers are important and should be addressed.

8.4 Takeaway

Privatization is neither pure efficiency nor pure redistribution. In Mexico it generated both.